

to 13B) is beneficial across tasks. Taken together, this indicates that proxy-tuning combines the benefit of both larger pretraining scale and task-specific tuning.

For GSM, proxy-tuned models follow the strict formatting of the task data, which are seen only by the task expert (see Appendix E for examples). For instance, 99.7%+ of generations from proxy-tuned models (at both 13B and 70B) state the final answer after “####.” Thus, proxy-tuning can promote even extremely unlikely tokens to the top of the probability distribution, enabling pretrained models to learn originally unlikely task formats.

6 Analysis

Using the instruction-tuning setup from §3, we analyze how proxy-tuning operates at the token level (§6.1) and whether the strength of tuning can be controlled via a new hyperparameter (§6.2).

6.1 What kinds of tokens are most influenced by proxy-tuning?

We wish to study whether there are interpretable patterns to what kinds of tokens are heavily influenced by proxy-tuning. To do this, we record the next-token probability distribution at each time step both from 13B-BASE and its proxy-tuned version. Then we take the difference in probabilities Δ_t assigned to the top token x_t chosen by the proxy-tuned model $\tilde{\mathcal{M}}$. I.e.,

$$\Delta_t = p_{\tilde{\mathcal{M}}}(x_t | x_{<t}) - p_{\mathcal{M}}(x_t | x_{<t}) \quad \text{where } x_t = \operatorname{argmax}_{\tilde{\mathcal{M}}}(X_t | x_{<t})$$

For GSM, we specifically compare Δ_t for tokens on the left-hand side (LHS) of intermediate equations, which requires formulating the correct reasoning, and those on the right-hand side (RHS), for which there is a single correct answer. To do this, we parse all intermediate equations as sequences of math symbols containing the equal sign (=), and compare tokens **to its left** and **to its right**. An example parse is shown in Table 1.

We find that Δ_t is 0.131 on average for LHS tokens and 0.056 for RHS tokens (a difference that is statistically significant with a p-value < 0.0001 under a t -test), suggesting that proxy-tuning contributes more to formulating reasoning steps than generating factual statements.

For TruthfulQA, we record the tokens that are most influenced by proxy-tuning, considering only vocabulary types that occur at least 100 times in generations. In Table 6, we show the 10 types whose probability increased the most from LLAMA2-13B to its proxy-tuned version, along with the 4-grams that they most commonly appear in as an example context. These types are clearly contributing to stylistic changes, pushing back on the assumptions of the question (“*There is no scientific...*”), pointing out common misconceptions (“*is a common myth*”), refraining from answering (“*I cannot provide*”), and acknowledging the complexity of the issue (“*depending on several factors*”).

Overall, these findings are consistent with the hypothesis that instruction-tuning mainly influences reasoning and style, rather than increasing the model’s knowledge.

6.2 Can a hyperparameter provide more granular control over steering?

Next, we explore the impact of introducing a hyperparameter to the proxy-tuning formula in Eq. (1) to control the amount of modification to $s_{\mathcal{M}}$, as follows: $s_{\mathcal{M}} + \alpha(s_{\mathcal{M}+} - s_{\mathcal{M}-})$. Intuitively, larger α magnifies the contrast between the expert and anti-expert, whereas smaller α leads to more similar predictions to the original base model. Note that this hyperparameter was introduced in the DEXPERTS paper, but we leave it out of our main experiments for simplicity.

We show the results of $\alpha \in [0.2, 2.0]$ for TruthfulQA in Table 2, where generations are evaluated on the axes of both informativeness and truthfulness. For truthfulness, increasing α leads to consistent gains, potentially because instruction-tuning improves a model’s commitment to factuality in the face of misleading questions. In contrast, the informativeness peaks at $\alpha = 0.4$, perhaps because while some instruction-tuning helps models address the

Token	Top Context
Here	Here are some of
Additionally	Additionally, it is important
There	There is no scientific
While	While some people may
several	depending on several factors
It	It's important to
provide	I cannot provide
respect	is important to respect
common	is a common myth
personal	I don't have personal

Table 6: For TruthfulQA, the 10 tokens whose probability increased the most from LLAMA2-13B to its proxy-tuned version. **Top Context** shows the most common 4-gram that the word occurs in, for example context.

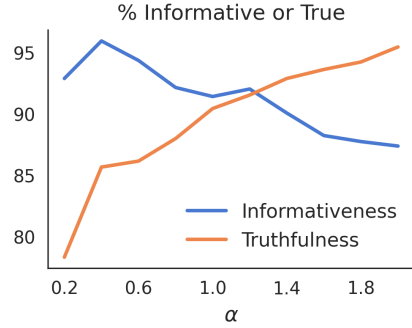


Figure 2: The percentage of responses to TruthfulQA which are informative or true, when varying α in $s_{\mathcal{M}} + \alpha(s_{\mathcal{M}^+} - s_{\mathcal{M}^-})$. We observe that α smoothly trades off between these two metrics.

question, excessive tuning increases the tendency to decline to answer. Overall, the smooth tradeoff indicates that α can be adjusted by a user depending on their application.

7 Case Study: Proxy-Tuning GPT-3.5 for the Present

One long-standing challenge for LMs is that they become outdated as the world evolves, which can be combated through continued pretraining on more recent data (Lazaridou et al., 2021; Onoe et al., 2022; Luu et al., 2022). In this case study, we apply proxy-tuning to temporally adapt a truly black-box LM, GPT-3.5 (specifically gpt-3.5-turbo-0613), whose training data is reported as stopping at September 2021. In this setting, we have extremely coarse information about the base model’s predictions, as the API provides log probabilities for *only the top 5 tokens*.⁴ So we consider a multiple choice (MC) setting where there are only four tokens of interest, namely the options {A, B, C, D}. We use **REALTIMEQA** (Kasai et al., 2023), which is updated periodically with questions from news sites. At the time of download, the dataset contains 3,531 examples.⁵

To obtain the expert, we update LLAMA2-7B on more recent data. In a realistic setting we would use web-scraped data from after 2021, but for the sake of compute, we instead mimic this by training an “oracle” expert on only relevant data. Specifically, we use the Google API to retrieve 10 articles for each query in REALTIMEQA, and continue pretraining LLAMA-7B on the retrieved articles.

For evaluation, we consider the model prediction to be the highest-probability token in {A, B, C, D}. We exclude any questions for which all answer choices are missing (only 1.8% of questions). Proxy-tuning only reweighs the four tokens of interest.

Results are shown in Table 7. Note that GPT-3.5 performs substantially better than random at 54.2% — outperforming even the small expert by a large margin — perhaps because some questions have guessable answers or due to more recent instruction-tuning data. Nonetheless, proxy-tuning improves the accuracy of GPT-3.5 by 2.3%, a statistically significant difference under a t -test with $p < 0.0001$. Thus while the expert and anti-expert are both weaker than GPT-3.5, contrasting their predictions yields a positive signal for the base model, representing an instance of weak-to-strong generalization (Burns et al., 2023).

Model	Acc.
<i>Llama2 7B</i>	
Base	28.4
Directly tuned	37.2
<i>GPT-3.5</i>	
Base	54.2
Proxy-tuned	56.5

Table 7: Results of proxy-tuning GPT-3.5 on REALTIMEQA.

⁴Note that the API also does not allow conditioning on partial model responses; it always generates the start of a new conversational turn. This prevents us from applying proxy-tuning to any task involving generation of more than one token.

⁵We use data from Jun 13, 2022 (the start of the dataset) to Dec 1, 2023.

8 Related Work

Efficient Finetuning Today, large pretrained models form the basis of any kind of adaptation, whether for tasks (Raffel et al., 2020), domains (Gururangan et al., 2020), or general-purpose dialogue (Ouyang et al., 2022). Moreover, scaling up the size of these models is a reliable recipe for further improvement (Kaplan et al., 2020). Thus, efficiently tuning ever-larger models has become a pressing challenge, leading to a large body of work on efficient finetuning, commonly through updating a small number of parameters (Houlsby et al., 2019; Li & Liang, 2021; Hu et al., 2022; Dettmers et al., 2023, i.a.). Nonetheless, these methods require white-box model access, which is unavailable for many of today’s advanced models.

In this context, “tuning” LMs at decoding-time represents an approach for efficient finetuning. Our work shares a similar vision with contemporary work (Mitchell et al., 2024), which applies the same DEXPERTS equation as operationalized in §3. However, they mainly view the equation as a tool for disentangling the effects of scaling up pretraining versus instruction-tuning, and do not measure the method’s effectiveness on existing benchmarks. In contrast, our work demonstrates the empirical strength of proxy-tuning, as well as its generality beyond instruction-tuning alone. Recently, Ormazabal et al. (2023) also combine the probability distributions from a small tuned model and a large pretrained model, but through a learned combination function which requires additional data and training.

For instruction-following specifically, a curated prompt can elicit generations that are surprisingly competitive with instruction-tuning (Han, 2023; Lin et al., 2023). However, these prompts tend to be quite long, introducing an inference-time computational burden and restricting the length of generations for models with limited context windows.

Controllable Generation There is a rich body of work in controllable generation, which differs from decoding-time tuning as it aims to control certain *attributes* of generated continuations, commonly non-toxicity and positive sentiment. In this space, there are many methods that operate on output logits (Krause et al., 2021; Liu et al., 2021; Yang & Klein, 2021; Deng & Raffel, 2023). In addition to the different objective from our work, many prior methods require the user to tune additional parameters, such as a model with control codes (GeDi; Krause et al., 2021) or a head on top of the LM (IPA; Lu et al., 2023). In contrast, proxy-tuning allows users to leverage the rich collection of small tuned models available online, potentially composing them off-the-shelf with no additional training.

Logit Arithmetic Our work builds off DEXPERTS (Liu et al., 2021), which introduced Eq. (1) and showed the effectiveness of ensembling logits from multiple LMs, an idea which was also briefly explored in earlier work (Dou et al., 2019). There has been a growing body of methods that perform arithmetic on multiple logit distributions for better text generation, such as contrasting the logits of a large and small model (Li et al., 2023), logits from different *layers* of a model (Gera et al., 2023; Chuang et al., 2023), and logits from the same model given different inputs (Shi et al., 2023; Pei et al., 2023; Sennrich et al., 2023; Leng et al., 2023); it has even been extended to non-autoregressive LMs (e.g., diffusion LMs; Han et al., 2024).

9 Conclusion

Proxy-tuning enables “tuning” of large language models at decoding-time by modifying output logits. It increases the accessibility of large LMs for those who lack the extensive resources required to train them, and addresses an important issue about how to adapt proprietary models to diverse use cases. At a minimum, we encourage model-producing organizations to share output probabilities from their models to enable use of these methods.

Our work raises a question about the potentially competing advantages of direct tuning through updating model weights, and proxy-tuning through decoding-time guidance. Indeed, full finetuning is an invasive approach that risks forgetting of previously learned information (McCloskey & Cohen, 1989); for instruction-tuning, this has sometimes been dubbed the “alignment tax” (Ouyang et al., 2022). We hope that proxy-tuning is a first step toward further exploration of customizable, algorithmic, decoding-time tuning.